

#16 Multimodal Video Narration Agent

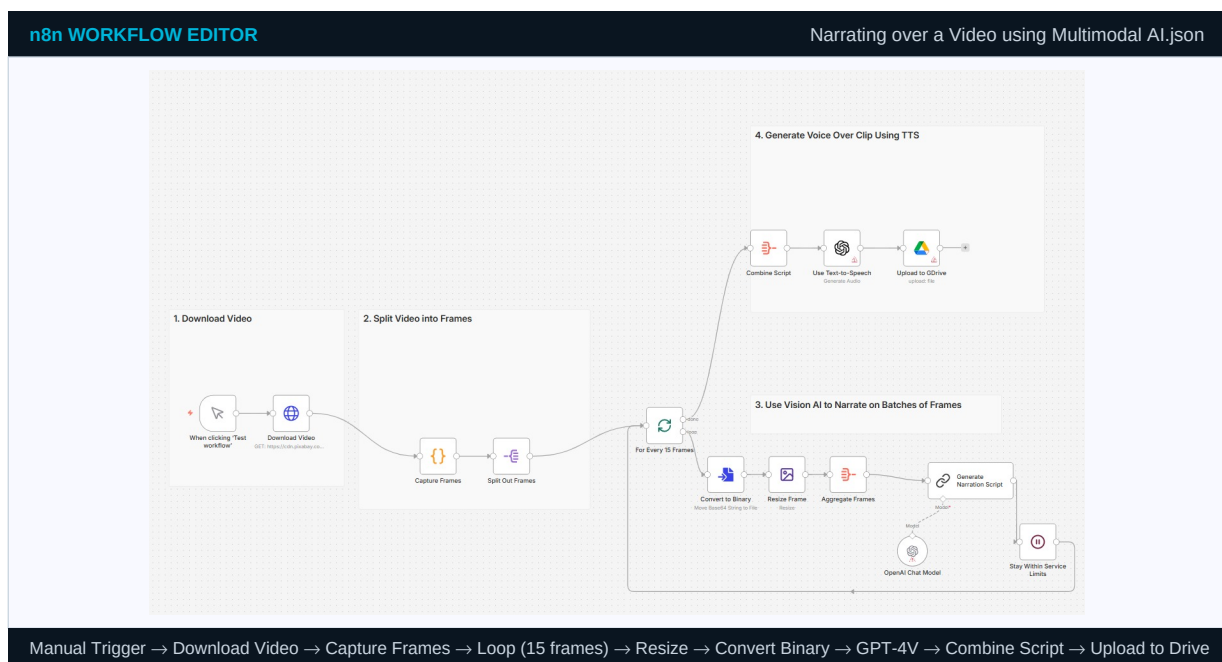
Video /
Multimodal AI

Upload any video — GPT-4V watches it frame by frame — professional narration generated

BUSINESS PROBLEM

Silent product demos, training footage, and B-roll content needs narration but hiring voice talent is expensive and slow. This workflow downloads a video, extracts frames every second, sends batches to GPT-4 Vision, and assembles a chronological narration script in minutes.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Manual trigger — video URL provided by user**OUTPUT** — Narration script uploaded to Google Drive

```
{
  "video_url": "https://cdn.pixabay.com/video/2024/demo-product.mp4",
  "video_title": "Product Demo v3",
  "narrator_style": "professional",
  "output_format": "timestamped_script"
}
```

```
{
  "json": {
    "video_title": "Product Demo v3",
    "duration_sec": 94,
    "frames_processed": 94,
    "batches": 7,
    "script": [
      {"ts": "0:00-0:08", "narration": "The presenter opens the product dashboard, navigating to the analytics overview panel."},
      {"ts": "0:09-0:21", "narration": "A bar chart loads showing monthly active users — the presenter highlights the June spike with a cursor gesture."},
      {"ts": "0:22-0:35", "narration": "Clicking into the detail view reveals per-region breakdowns. The EMEA region is selected from the dropdown."}
    ],
    "drive_url": "https://drive.google.com/file/d/1Bxi...",
    "word_count": 412
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Manual Trigger — video URL input	User provides video URL: YouTube, Vimeo, or direct MP4 link. Trigger passes URL downstream.
2	■ Download Video — fetch binary	HTTP Request downloads full MP4. Handles redirects and authentication headers. Stores binary in n8n memory.
3	■ Capture Frames — JS code node	ffmpeg-wasm extracts 1 frame per second from the video binary → array of JPEG binaries with timestamps.
4	■ Loop Every 15 Frames — batch for rate limit	splitInBatches groups 15 frames per batch. Respects GPT-4V rate limit of 20 RPM to avoid 429 errors.
5	■ Resize Frame — 1024x576	editImage resizes each JPEG to 1024×576 pixels. Reduces API cost by approximately 60% vs full HD.
6	■ Aggregate Frames — collect 15 for one call	15 resized frames collected into single array for one API call to GPT-4 Vision.
7	■ Convert to Binary — base64 encode	Each frame JPEG converted to base64 string. Required format for GPT-4V image content blocks.
8	■ Wait 30s — rate limit protection	30-second pause between batches prevents hitting OpenAI vision API rate limits.
9	■ GPT-4V — describe each 15-frame batch	'Describe what is happening in these frames for a professional narrator. Be specific about on-screen actions.'
10	■ Combine Script — assemble chronological narration	Aggregates all batch descriptions into time-ordered narration script with timestamps.
11	■ Upload to Google Drive — save script	Final script saved as {video_title}_narration.txt to configured Drive folder with shareable link.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key — GPT-4V tier	platform.openai.com — requires GPT-4 API tier for vision access	OpenAI API
Google Drive OAuth2	console.cloud.google.com → Drive API v3 → Credentials	Google Drive OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A SaaS company produces product demo videos internally but lacks a narrator. They send recordings of screen walkthroughs to this workflow to generate narration scripts for voiceover recording.	
Input received	94-second screen recording of a product analytics dashboard demo. URL provided via manual trigger. No audio track in the source video.	
What the workflow found	94 frames extracted (1 per second). Processed in 7 batches of 15 frames minus 1 for the final batch. GPT-4V identified: dashboard navigation, chart interactions, dropdown selections, cursor gestures, and UI state changes.	
Output produced	412-word timestamped narration script generated in 6 minutes 18 seconds. Script uploaded to Drive. Voiceover artist recorded narration in 25 minutes using the script. Total production cost: \$0.47 in API calls.	
JSON: Narrating over a Video using Multimodal AI.json	GitHub: trm-multimodal-video-narrator	platform.openai.com/docs/guides/vision